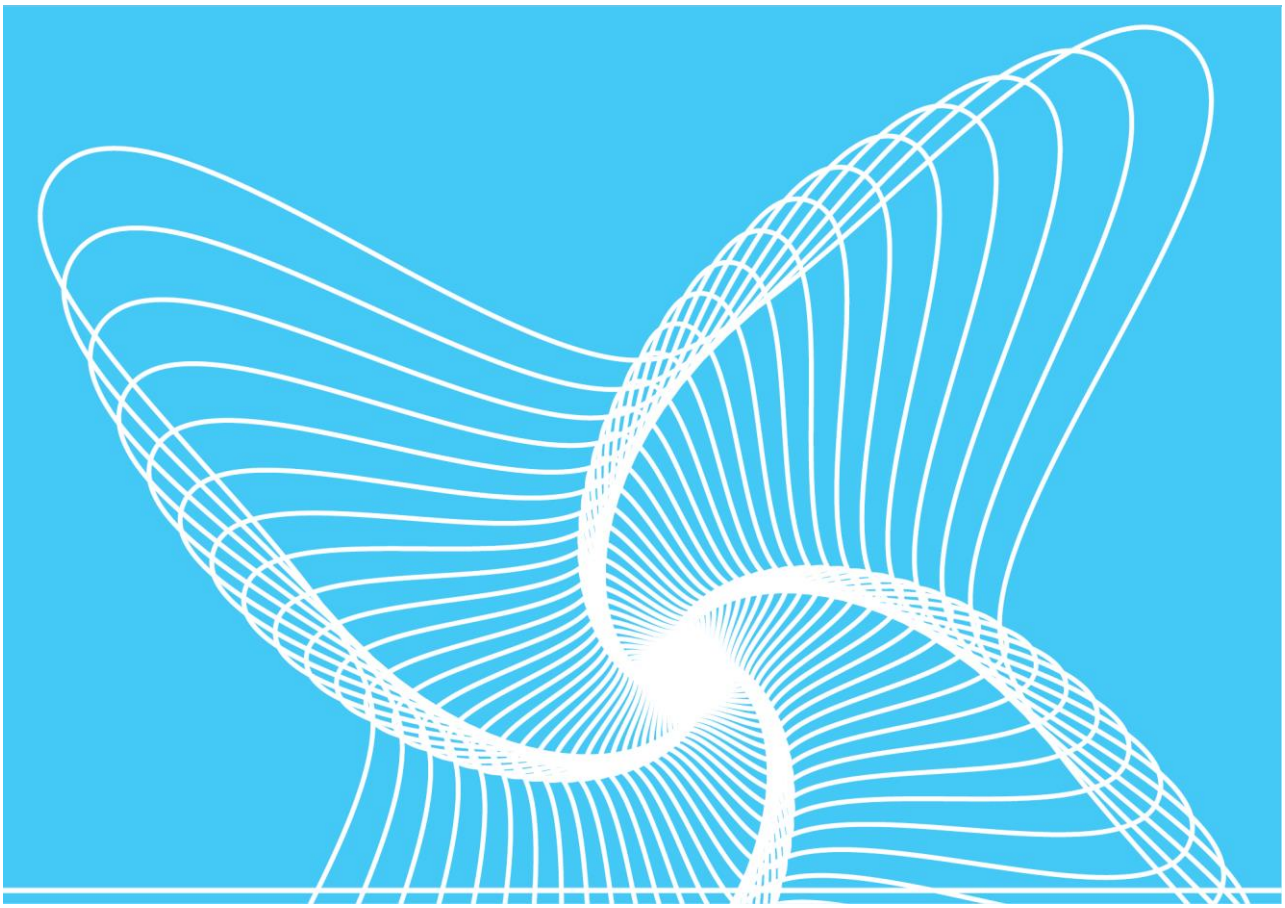
DNV·GL

SESAM USER MANUAL

# StableLines

## Pipeline Tools application for on-bottom stability design of submarine pipelines according to DNVGL-RP-F109

Valid from program version 1.7



SAFER, SMARTER, GREENER

Sesam User Manual

StableLines

Date: 13 April 2018, Revision 00

Valid from StableLines version 1.7

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# 1 INTRODUCTION

## 1.1 StableLines – On-bottom stability design of submarine pipelines

StableLines is a Microsoft Excel VBA spreadsheet developed by DNV GL for design and assessment of on-bottom stability for submarine pipeline in compliance with DNVGL-RP-F109 "On-Bottom Stability Design of Submarine Pipelines", issued May 2017[1]. DNVGL-RP-F109 harmonizes with the design method in DNVGL-ST-F101 "Submarine pipeline systems", ref. /3/.

## 1.2 How to read the Manual

- Read section 2 to learn about the features of StableLines.
- Read section 3 to learn the basics of using StableLines and program limitations.
- Read section 4 for a detailed description of the user interface, learn how to use StableLines and alternative ways of running the program.

## 1.3 Status List

In addition to this User Manual you may find additional information in the Status List (as for any other SESAM PIPELINE program). Such information may be:

- Reasons for update (new version)
- New features
- Errors found and corrected

Access this information by clicking the customer login link in our website www.dnvgl.com/software/.

## 1.4 Acronyms and symbols frequently used in the User Manual

The table below lists a few acronyms and symbols that are frequently used in the User Manual. Reference is also given to DNVGL-RP-F109, ref. /1/.

| | |
|---|---|
| $C_Y$* | Peak horizontal load coeffcient |
| $C_Z$* | Peak vertical load coeffcient |
| *d* | Water depth |
| *FJC* | Field Joint Coating |
| $F_R$ | Passive soil resistance |
| $F_Y$ | Horizontal hydrodynamic (drag and inertia) load |
| $F_Z$ | Vertical hydrodynamic (lift) load |
| *Hs* | Significant wave height during a sea state |

[1] The May 2017 issue of DNVGL-RP-F109 was merely a re-branding of the RP. Its design methodologies thus fully align with DNV-RP-F109 issued October 2010, ref. /2/.

| | |
|---|---|
| *K* | Keulegan Carpenter number |
| *L* | Significant weight parameter |
| *M* | Steady to oscillatory velocity ratio |
| *N* | Spectral acceleration factor |
| *SI* | Système International d'Unités |
| *OD* | Outer Diameter |
| $T_p$ | Peak period for design spectrum |
| $T_u$ | Spectrally derived mean zero up-crossing period |
| $U_s$ | Spectrally derived oscillatory velocity (significant amplitude) for design spectrum, perpendicular to pipeline |
| *V* | Steady current velocity associated with design spectrum, perpendicular to pipeline |
| *VBA* | Visual Basic for Application |

# 2 FEATURES OF STABLELINES

StableLines is a Microsoft Excel Visual Basic (VBA) based spreadsheet that incorporates the design principles of DNVGL-RP-F109, ref. /1/, for on-bottom stability design of submarine pipelines.

StableLines v1.7-01 provides the following features:

- StableLines includes design modules for both pipelines and umbilicals.
- Pipeline dimensions and weights are automatically calculated by defining thicknesses, cut back length (field joint coating (FJC)) and densities for the different layers: Steel, Coating, Marine growth and FJC.
- Stable Umbilical weight is automatically calculated based on outer diameter.
- Two different criteria for calculating stability:
    - Absolute lateral static stability method
    - Generalized lateral stability method with displacements up to 10 OD (Outer Diameter)
- Four different pre-defined soil models: Sand, Clay, Rock and Coulomb friction (user defined).
- Short term, stationary and irregular sea states are modelled based on the JONSWAP spectrum.
- Wave induced particle velocities at seabed level is automatically calculated based on user inputs significant wave height (*Hs*) and peak period (*Tp*).
- Optional reduction of current velocity due to boundary layer near the seabed.
- Reduction of both current and wave induced velocities based on directionality with independent specification of wave and current directions.
- Possibility of user-defined reduction factor due to added seabed penetration of Pipeline/Umbilical.
- Pipeline in empty condition and in operational condition are calculated simultaneously.
- StableLines suggests added steel and concrete/armour thicknesses to make unstable Pipelines stable.
- Iteration module calculates optimal necessary pipeline weight and concrete thickness.
- Graphical presentations of:
    - Required submerged weight as function of direction.
    - Required concrete thickness as function of concrete density (pipeline mode) or required weight as a function of outer diameter (umbilical mode).
    - Required submerged weight as function of current velocity.
    - Dimensionless weight parameter *L* as function of *K* (Keulegan-Carpenter number) and sensitivity to *M* (ratio of current velocity to wave induced particle velocity).
- Report of final and intermediate results ready for "copy-paste" into Technical reports.
- Powerful tool to analyse stability of a large number of cases simultaneously (Parametric Runs).

# 3 GETTING STARTED

## 3.1 Licensing

StableLines v1.7 is licensed through FlexLM License Manager. A valid license file is required to run StableLines v1.7. Please contact Software.Support@DNVGL.com if any questions.

## 3.2 User Interface

StableLines is a Microsoft Excel Visual Basic (VBA) program. it is based on Microsoft Excel 2016 and the user interface is the same as in a typical Excel spreadsheet with direct input into cells, push buttons and pull-down menus.

The program has been designed so that the input and results are located on the Main sheet. The user can see all the necessary information from this sheet without navigating between windows. There are four kinds of sheets available within StableLines:

- Main sheet - contains all the important input and output data.
- Report sheet – contains results and useful intermediate calculations that may be printed and used as appendix sheet directly in reports etc.
- Plots sheet – contains results for graphical presentation and additional useful results.
- Parametric runs sheet – used to run several on-bottom stability cases simultaneously (this sheet is created when the "Make new Parametric runs sheet" button on the Main sheet is pushed).

The user can copy the Parametric runs sheet, but the program can only operate with a single Main sheet.

The sheets have been designed so that the user can modify input cells only. This is to prevent corruption of the input data, formulae and the routines. Cells with input access are white, whereas cells that do not allow user access are coloured light yellow. Some of the cells change access rights according to specific selections. This occurs when some of the User Options have been changed.

## 3.3 Performing an analysis

The procedure to follow when assessing a single on-bottom stability case is briefly described below:

1. Give input data in the Main sheet:
    a. Pipeline and coating data, or Umbilical data
    b. Soil interaction data
    c. Environmental parameters
2. Enter the relevant wave and current data, including relevant directions considered.
3. Choose soil type.
4. Choose stability criterion.
5. Choose design condition(s).
6. Push Calculate button in the Main sheet to perform calculations and update on-bottom stability results, including graphs.
7. For Pipelines: update either steel or concrete thicknesses with the built-in buttons and re-run calculations in order to find optimized required weight.

The results are presented on the Main sheet in terms of submerged weights and necessary concrete/armouring and steel thicknesses for both empty and operational conditions for pipeline.

In Umbilical mode, only the required weight for achieving lateral stability is given.

## 3.4 Warnings and Errors

When the calculate button is clicked, StableLines read input parameters and perform a validation check on these. If a parameter is outside its validity range, according to DNVGL-RP-F109, a Warning or Error message pops up providing information on what parameter that is not valid and thus should be changed.

StableLines differs between two types of messages to the user:

- **Warning** message: The calculation is in some way outside validity ranges governed by DNVGL-RP-F109, however the calculation is completed and an indicative result is provided.
- **Error** message: The calculation is in some way outside validity ranges governed by DNVGL-RP-F109 or there exist a false input value, and the calculation is terminated.

## 3.5 Getting help

Comments are provided in many of the cells to give further guidance to the user. The comments give additional definitions and references to DNVGL-RP-F109. Comments are identified by a small red triangle in the top right-hand corner of the cell and are viewed by pointing at the cell.

## 3.6 Limitations of StableLines

### 3.6.1 Limitation to design parameters

StableLines will give warnings when limitations set by DNVGL-RP-F109 are violated, ref. Section 3.4. Limitations that most frequently generates questions from users are related to friction coefficient, trench depth, specific gravity and shallow waters.

Note also that end constraints, i.e. effects of tension and bending moments due to tie-in points, are not handled by StableLines.

### 3.6.2 On alignment with DNVGL-RP-F109

As stated in Section 2, StableLines incorporates the design principles of DNVGL-RP-F109, ref. /1/. However, some features not yet officially recognized in the RP have already been incorporated in StableLines. These are:

- A cut-off at $M$ = 0.2 is introduced for the 0.5D stability criterion on clay (equation 3.36 in DNVGL-RP-F109), as the existing equation give un-physical large weight for low values of M. (*since v1.5*)
- Equation 3.35 is adjusted to account for that the design tables for 10D criterion on sand where calibrated for a design storm of 1000 waves, while for clay the design tables where calibrated for a design storm of 3 hours duration. The existing equation 3.35 indicates that design tables for both sand and clay were calibrated for a design storm of 1000 waves. (*since v1.4*)
- The criterion that $G_c$ must be ≤ than 2.78 for the Generalized Method has been removed. It is shown that when $G_c$ > 2.78 the sensitivity of $w_{s,required}$ for varying $G_c$ is very low due to very stiff clay for $G_c$ ≥ 2.78. (*since v1.4*)
- For the absolute stability criterion, the peak horizontal load $F_Y^*$ is calculated by the equation below when $K^*$ < 2.5:
  $F_Y^* = r_{tot,y} \cdot \frac{1}{2} \cdot \rho_w \cdot D \cdot \left( \frac{C_M \cdot \pi^2}{K^*} \cdot U^{*2} + C_Y^* \cdot (U^* + V^*)^2 \right)$,
  where $C_M^*$ = 3.29 and $C_Y^*$ = 1.00. This is based on an FAQ to DNV-RP-F109. (*since v1.3, corrected in v1.7*)

All these features are expected to be incorporated into the next release of DNVGL-RP-F109.

### 3.6.3 On umbilicals, cables and small diameter pipelines

The basic approach for the generalized method in DNVGL-RP-F109 builds on experiments and analyses performed in the 1980s prior to issuing DNV-RP-E305 in 1988. To DNV GL's knowledge, the experiments that were performed to establish load – and soil models were performed on cylinders with a diameter range of 300 mm to 600 mm. From a strict scientific point of view, this is thus the applicability range for the method. However, this approach, i.e. DNV-RP-E305 and DNVGL-RP-F109, has been used for pipelines with much larger diameters for almost 30 years and has proven to be a sound approach. The experience with smaller diameter umbilicals and cables is much shorter, but DNV GL has no reason to believe that the approach should not be sufficiently conservative for smaller diameter pipelines, umbilicals and cables.

The absolute stability criterion did not exist in DNV-RP-E305, but the models rely on the same experiments, so what is stated above is also applicable to this method.

Finally, it should be noted that both methods leads in general to much higher specific gravity, i.e. weight divided by volume, for small diameters which also indicates that the methods are valid for these cases.

### 3.6.4 On Linear wave theory

Note that the calculation of significant wave velocity $U_s$ and mean zero up-crossing period $T_u$ in StableLines are based on linear wave theory, as described in DNVGL-RP-F109, Ref. /1/. Some combinations of return period values for significant wave height $H_s$, peak period $T_p$ and water depth $d$ may represent sea states that are not adequately described by linear wave theory. Whether linear wave theory is applicable for the given input data must be validated by the user. Some guidance on this matter may be found in DNV-RP-C205, ref. /4/.

# 4 EXECUTION OF STABLELINES

## 4.1 Main sheet input

### 4.1.1 General

The main structure of the Main sheet of StableLines is described in Figure 4-1 below.

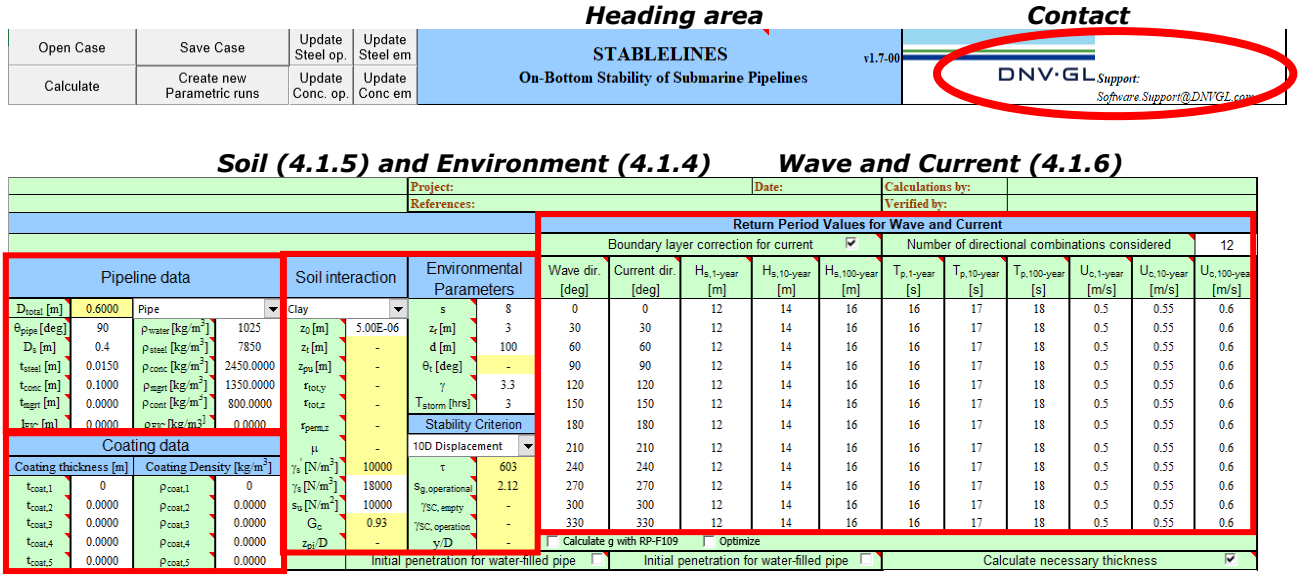



**Figure 4-1 Main sheet structure**

All input and output parameters presented in the main sheet are introduced with SI units. In addition, the StableLines main sheet contains a comprehensive list of comments describing the various parameters entering the analysis. The comments are shown when the cursor is held over the cells with a red triangle in the upper right corner.

### 4.1.2 Buttons

All main sheet buttons are located in the upper left corner of the heading area. Below follows a short description of button functionality.

*Calculate* The calculation routine in the Main sheet is initiated using the Calculate button. To update the main sheet results, the Calculate button has to be clicked.

*Save case* This button saves the actual case with all input data presented in the main

sheet as a .txt file (StableLines Case file).

*Open case* This button reads input from a StableLines Case file and presents it in the Main sheet. To retrieve the results for this case, press the Calculate button.

*Create New Parametric runs* This button makes a clean Parametric runs sheet. This sheet may be used to perform sensitivity analyses of parameters etc. For keeping several parametric run cases, rename the sheet from Parametric runs to another name. For further description of the usage of this sheet, see Section 4.3.

*Update steel op* This button updates the steel thickness and outside diameter with the steel thickness necessary for achieving a stable pipeline during operation, ref. Section 4.2.1. This requires that a calculation has been performed to have a result to use for update. Similar buttons also exists for empty condition for steel and both operation and empty condition for concrete thickness. These buttons do not have any function in umbilical mode.

## 4.1.3 Pipeline/Umbilical and Coating Data

The user should enter all pipeline and coating details into this section, see Figure 4-2 below. The input data can be either for Pipeline or Umbilical mode. Typical Pipes and umbilical geometries are shown in Figure 4-2.

| Pipeline data | | | |
|---|---|---|---|
| $D_{total}$ [m] | 0.6000 | Pipe | |
| $\theta_{pipe}$ [deg] | 90 | $\rho_{water}$ [kg/m$^3$] | 1025 |
| $D_s$ [m] | 0.4 | $\rho_{steel}$ [kg/m$^3$] | 7850 |
| $t_{steel}$ [m] | 0.0150 | $\rho_{conc}$ [kg/m$^3$] | 2450.0000 |
| $t_{conc}$ [m] | 0.1000 | $\rho_{mgrt}$ [kg/m$^3$] | 1350.0000 |
| $t_{mgrt}$ [m] | 0.0000 | $\rho_{cont}$ [kg/m$^3$] | 800.0000 |
| $l_{FIC}$ [m] | 0.0000 | $\rho_{FIC}$ [kg/m3] | 0.0000 |
| **Coating data** | | | |
| Coating thickness [m] | | Coating Density [kg/m$^3$] | |
| $t_{coat,1}$ | 0 | $\rho_{coat,1}$ | 0 |
| $t_{coat,2}$ | 0.0000 | $\rho_{coat,2}$ | 0.0000 |
| $t_{coat,3}$ | 0.0000 | $\rho_{coat,3}$ | 0.0000 |
| $t_{coat,4}$ | 0.0000 | $\rho_{coat,4}$ | 0.0000 |
| $t_{coat,5}$ | 0.0000 | $\rho_{coat,5}$ | 0.0000 |

| Umbilical data | | | |
|---|---|---|---|
| $D_{total}$ [m] | 0.1200 | Umbilical, submerged | |
| $\theta_{pipe}$ [deg] | 90 | $\rho_{water}$ [kg/m$^3$] | 1025 |
| $D_o$ [m] | 0.12 | | |
| | | $\rho_{mgrt}$ [kg/m$^3$] | 1350.0000 |
| $t_{mgrt}$ [m] | 0.0000 | | |
| **Coating data** | | | |
| Coating thickness [m] | | Coating Density [kg/m$^3$] | |
| $t_{coat,1}$ | - | $\rho_{coat,1}$ | - |
| $t_{coat,2}$ | - | $\rho_{coat,2}$ | - |
| $t_{coat,3}$ | - | $\rho_{coat,3}$ | - |
| $t_{coat,4}$ | - | $\rho_{coat,4}$ | - |
| $t_{coat,5}$ | - | $\rho_{coat,5}$ | - |

**Figure 4-2 Pipeline and Coating input data for Pipeline mode and Umbilical mode.**

**Pipeline data**

$D_{total}$ total outer diameter of the pipeline (including all layers)

$\theta_{pipe}$ pipeline heading relative to geographic north

$D_s$ outer steel diameter for pipeline

$t_{steel}$ steel thickness

| | |
|---|---|
| $t_{conc}$ | thickness of concrete coating |
| $t_{mgrt}$ | thickness of marine growth |
| $l_{FJC}$ | cut back length of the field joint coating (taken as the length from parent coating to line pipe end, i.e. *half* the distance between two adjacent parent coatings.) Note that the line pipe length of 12.2 m is incorporated without possibility to change. |
| $\rho_{water}$ | density of seawater surrounding the pipeline |
| $\rho_{steel}$ | density of steel |
| $\rho_{conc}$ | density of concrete coating |
| $\rho_{mgrt}$ | density of the marine growth |
| $\rho_{cont}$ | density of the content inside the pipeline in operational condition. |
| $\rho_{FJC}$ | density of the field joint coating |

Note that marine growth is not included in the calculations for pipeline in empty condition.

**Umbilical data**

| | |
|---|---|
| $D_{total}$ | is the total outer diameter of the pipeline (including all layers) |
| $\theta_{pipe}$ | is the pipeline heading relative to geographic north |
| $D_o$ | is the outer diameter of the umbilical, note: *not* including marine growth |
| $t_{mgrt}$ | is the thickness of marine growth |
| $\rho_{water}$ | is the density of seawater surrounding the umbilical |
| $\rho_{mgrt}$ | is the density of marine growth |

**Coating data (Pipeline mode only)**

Five layers of external coating can be introduced into the calculations. These coating layers are specified with their respective thicknesses and material densities. Coating data can only be specified in Pipeline mode.

**Cross Sections**

Calculations of submerged weights and necessary thickness of concrete and steel are performed using the pipeline and umbilical cross sections illustrated in Figure 4-3 below.

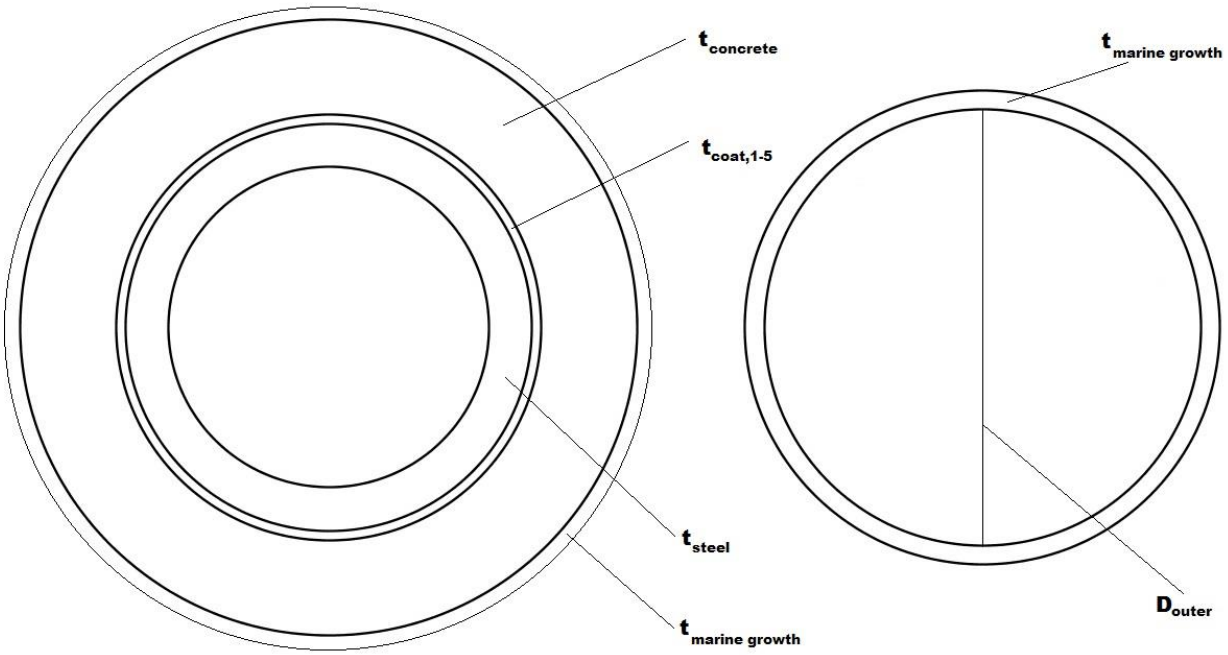



**Figure 4-3 Pipeline (left) and umbilical (right) cross sections used in StableLines**

The cross section used in Pipeline mode is shown to the left in Figure 4-3. When StableLines calculates necessary concrete thickness in order to obtain required pipeline weight, the concrete is added externally without affecting the outer steel diameter and other coating layers, except for marine growth where only the thickness is kept constant. StableLines also calculates the steel thickness that must be added to the original steel thickness for the submerged pipeline weight to match required submerged weight. In this case, the inner steel diameter and the *thickness* of external coating layers are kept constant, while the outer steel *diameter* is changed.

The cross section used in Umbilical mode is shown to the right in Figure 4-3. When the necessary weight is calculated in umbilical mode, the outer diameter is kept constant. If a layer of marine growth is specified in the analysis, the reported necessary weight is the submerged weight for the umbilical itself (not including weight of marine growth).

See also Section 4.2.1 for description of the output parameters $t_{conc,added}$ and $t_{steel,added}$.

### 4.1.4 Environmental Parameters and Stability criteria

The user should enter all environmental data, except for those parameters with light yellow background colour. These parameters are either given as helpful output or not used (-) in the calculations with the selected stability criterion.

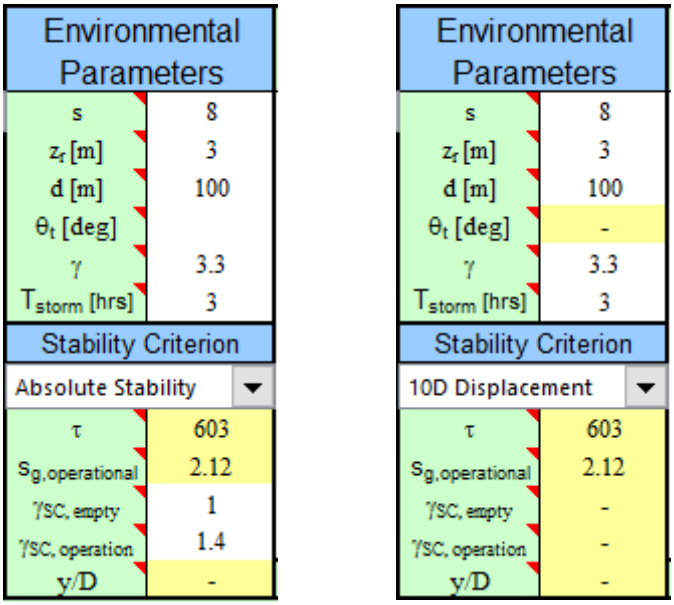



**Figure 4-4 Environmental parameters for the absolute stability criterion (left) and the generalized stability criteria (right)**

A pull down menu allow the user to choose from the following Stability criteria:

**Absolute Stability** no pipeline movement during the design storm

**.5D Displacement** 0.5 OD allowable movement during the design storm

**10D Displacement** 10 OD allowable movement during the design storm

**Intermediate** Between 0.5 OD and 10 OD allowable movement (specify *y/D*) during the design storm

The environmental and stability criterion parameters can be summarised as:

*s* Spectral spreading exponent used for wave reduction factor calculation

$z_r$ Reference height for current measurement

*d* Water depth

*θ* Trench wall angle (Absolute stability only)

*γ* Peak enhancement factor

$T_{storm}$ Storm duration specified in hours

*τ* Number of oscillations during the actual storm

$s_{g,operation}$ Specific gravity (initial weight for pipeline, final weight for umbilical), calculated for information

$\gamma_{SC,empty}$ Safety factor for empty condition used for Absolute stability

$\gamma_{SC,operation}$ Safety factor for operational condition used for Absolute stability

*y/D* Relative displacement limit for intermediate displacement criterion

In lieu of project specific data for the peak enhancement factor, $\gamma$, StableLines can calculate $\gamma$ according to equation (3.7) in /1/. This is done by checking the box shown in Figure 4-5. Note that $\gamma$ then will be calculated for every specified sea state.

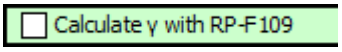



**Figure 4-5: Checkbox for automatically calculate γ using DNVGL-RP-F109.**

### 4.1.5 Soil Interaction

The user should enter all soil interaction data, except for those parameters with light yellow background colour. These parameters are either given as helpful output or not used (-) for the stability criterion and soil type chosen.

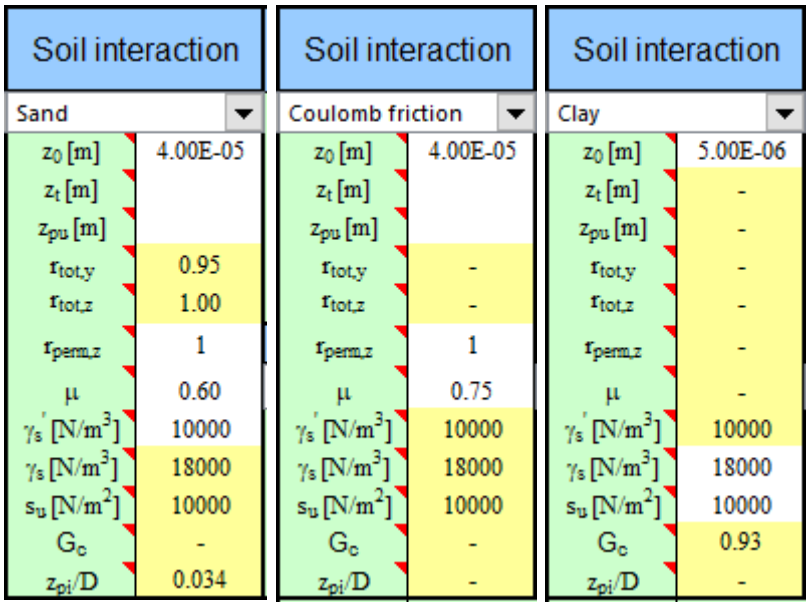



**Figure 4-6 Soil interaction for the stability criteria, absolute stability with sand (left), absolute stability with Coulomb friction (middle) and the generalized stability criterion (right).**

Depending on stability criterion chosen, there are different options in the soil interaction pull down menu.

**Absolute stability:**

- Sand
- Clay
- Rock
- Coulomb friction

Different friction factors are specified for the different soil types. The coefficient of friction is taken as 0.6 on sand, 0.2 on clay and 0.6 on rock. These are default values given by ref. /1/, but can be set lower. The Coulomb friction is given as input by the user, given as 0.75 in Figure 4-6.

**Generalized stability method:**

- Sand
- Clay

$\mu$ is not directly included in the calculations for generalized method, however values of 0.2 for clay and 0.6 for sand is used when establishing the results from fully dynamic analyses which the generalized method is based upon.

For the generalized stability method, the default sand and clay friction coefficients are implicitly applied in the DNVGL-RP-F109.

The soil interaction parameters can be summarised as:

| | |
|---|---|
| $z_0$ | Seabed roughness, depending on soil |
| $z_t$ | Trench depth, only valid for Absolute stability |
| $z_{pu}$ | Penetration depth, in additional to the initial penetration calculated directly in StableLines, from movement, piping, trench etc. The added penetration depth must however be calculated and validated by the user. |
| $r_{tot,y}$ | Total horizontal resistance factor, given as output to provide sensitivity check when changing $z_t$ and $z_p$. |
| $r_{tot,z}$ | Total vertical resistance factor, given as output to provide sensitivity check when changing $z_t$ and $z_p$. |
| $r_{perm,z}$ | Reduction factor on lift load due to permeable seabed. Only used for Absolute stability. |
| $\mu$ | Coefficient of friction |
| $\gamma_s'$ | Submerged unit soil weight (sand) |
| $\gamma_s$ | Dry unit soil weight (clay) |
| $s_u$ | Un-drained clay shear strength |
| $G_c$ | Soil strength parameter, informative output parameter only valid for soil type clay |
| $z_{pi}/D$ | Relative initial soil penetration for operation condition, the calculated value does not include $z_{pu}$ |

Initial penetration is calculated in order to determine passive soil resistance for Absolute stability assessments. When the boxes in Figure 4-7 are checked, the initial penetration will be calculated for pipeline filled with fresh water ($\rho_{water}$ = 1000 kg/m$^3$). The box for empty condition has been included for cases where the pipeline is installed water-filled. If the boxes are not checked, initial penetration is calculated for empty pipeline and by using the content density specified in the Pipeline data section for operational condition. The checkboxes are only active for the Absolute stability criterion in Pipeline mode.

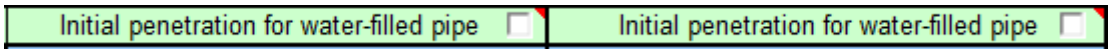



**Figure 4-7 Check-box for specifying whether initial penetration shall be calculated for water-filled pipeline during Absolute stability assessments.**

### 4.1.6 Return Period Values (RPV) for Wave and Current

The user should enter all Wave and Current details into this section, see Figure 4-8. All directions are defined as relative to geographic north. Up to 12 different directions can be specified.

Wave-induced flow and current can be specified independently in StableLines. The most conservative case is when wave-induced flow and current are assumed co-linear, i.e. they act in the same direction.

When using omni-directional values, the specified direction should be 90 degrees relative to the given pipeline direction.

Current velocities should always be specified as positive numbers. Directionality should be handled with the "Wave dir." and "Current dir." columns.

| Return Period Values for Wave and Current | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Boundary layer correction for current ☑ | | | | | Number of directional combinations considered | | | | | 12 |
| Wave dir. [deg] | Current dir. [deg] | $H_{s,1\text{-year}}$ [m] | $H_{s,10\text{-year}}$ [m] | $H_{s,100\text{-year}}$ [m] | $T_{p,1\text{-year}}$ [s] | $T_{p,10\text{-year}}$ [s] | $T_{p,100\text{-year}}$ [s] | $U_{c,1\text{-year}}$ [m/s] | $U_{c,10\text{-year}}$ [m/s] | $U_{c,100\text{-year}}$ [m/s] |
| 0 | 0 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 30 | 30 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 60 | 60 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 90 | 90 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 120 | 120 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 150 | 150 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 180 | 180 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 210 | 210 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 240 | 240 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 270 | 270 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 300 | 300 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |
| 330 | 330 | 12 | 14 | 16 | 16 | 17 | 18 | 0.5 | 0.55 | 0.6 |

**Figure 4-8 Wave and Current input for the relevant directions relative to geographic north.**

Return Period Values (RPV) for both empty and operational design conditions must be specified by the user, see Figure 4-9.

| Design condition for empty pipe | Design condition for pipe in operation |
|---|---|
| 1-year and 10-year RPV Combination | 1-year and 10-year RPV Combination |

**Figure 4-9 Design condition pull down menu**

A pull down menu allow the user to choose from the following Design conditions:

- **1-year and 1-year RPV Combination**
- **1-year and 10-year RPV Combination**
- **1-year and 100-year RPV Combination**
- **10-year and 10-year RPV Combination**
- **10-year and 100-year RPV Combination**
- **100-year and 100-year RPV Combination**

As specified in DNVGL-RP-F109, for permanent operational conditions and temporary phases with duration over 12 months, a 100-year return period applies. When detailed information about current and wave joint probability is not available, the RPV Combination 10-year and 100-year applies.

For temporary phases with duration less than 12 months and in excess of three days, a 10-year seasonal condition applies. For this case an approximate condition would be the RPV Combination 1-year and 10-year.

If different RPV values for wave and current are chosen, StableLines automatically reports the most conservative combination.

## 4.2 Viewing Results Output

### 4.2.1 Numerical Results

**Pipeline Mode**

The main results output are given by StableLines as shown in Figure 4-10 below for a pipeline case.

| Design condition for empty pipe | | Design condition for pipe in operation | |
|---|---|---|---|
| 1-year and 10-year RPV Combination | | 10-year and 100-year RPV Combination | |
| Results - Empty pipe | | Results - Pipe in operation | |
| $w_s$ [N/m] | 2329 | $w_s$ [N/m] | 3173 |
| $w_{s,\ required}$ [N/m] | 3602 | $w_{s,\ required}$ [N/m] | 4867 |
| $t_{conc,added}$ [m] | 0.038 | $t_{conc,added}$ [m] | 0.063 |
| $t_{steel,added}$ [m] | 0.014 | $t_{steel,added}$ [m] | 0.019 |
| $w_{s,\ vertical}$ [N/m] | 284 | $w_{s,\ vertical}$ [N/m] | 284 |

**Figure 4-10 Numerical results given in the Main sheet for Pipeline mode.**

The pull down RPV menu is taken as input during calculations, but do also work as an integrated part of the results section. The worst case of the 1-year wave, 10-year current and 1-year current, 10-year wave is presented for the empty pipeline. The worst case of the 10-year wave, 100-year current and 10-year current, 100-year wave is presented for pipeline in operation.

The main results can be summarised as:

| | |
|---|---|
| $w_s$ | Pipeline submerged weight, calculated from the user-specified pipeline data. |
| $w_{s,required}$ | Required submerged weight needed to achieve stable pipeline/umbilical for the lateral stability criterion chosen based on initial data.<br>Note: this value is not automatically optimized, as there are two routes of optimization: adjusted steel thickness, $t_{steel,added}$, and adjusted concrete thickness, $t_{conc,added}$. |
| $w_{s,vertical}$ | Necessary submerged weight to fulfil the vertical stability criterion in water (outer diameter = as specified). |
| $t_{conc,added}$ | Necessary added concrete thickness to fulfil the stability criterion. To find optimized submerged weight (i.e. $w_s = w_{required}$), update Pipeline data with |

the buttons *Update Conc. op/em*, ref. Section 4.1.2, and push the *Calculate* button again.

Calculation of $t_{conc,added}$ can be turned off by unchecking the "Calculate necessary thickness" checkbox.

$t_{steel,added}$ Necessary added concrete thickness to fulfil the stability criterion. To find optimized submerged weight (i.e. $w_s = w_{required}$), update Pipeline data with the buttons *Update Steel op/em*, ref. Section 4.1.2, and push the *Calculate* button again.

Calculation of $t_{steel,added}$ can be turned off by unchecking the "Calculate necessary thickness" checkbox.

To achieve a stable pipeline, $w_s$ should be equal to or larger than $w_{required}$ and $w_{s,vertical}$.

If the checkbox called "Optimize", see Figure 4-11, is checked off StableLines will calculate potential reduction in steel and concrete thicknesses if the pipeline is initially stable.

This checkbox will not have any function in the umbilcal mode.

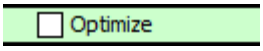



**Figure 4-11: Optimize checkbox to optimize the thickness of concrete and steel of an initally stable pipeline.**

### Umbilical mode

A different set of output is given in the Main sheet for Umbilical mode, see Figure 4-12.

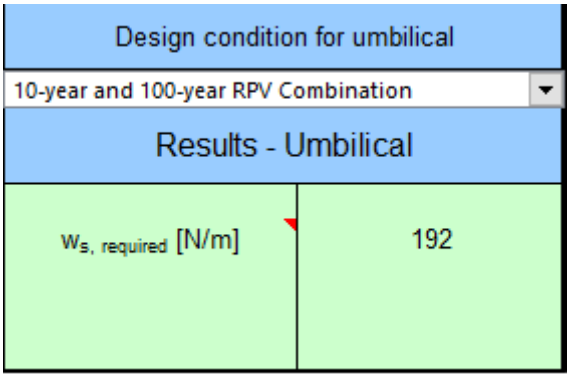



**Figure 4-12 Numerical results given in the Main sheet for Umbilical mode.**

The main results can be summarised as:

$w_{s,required}$ Required submerged weight needed to achieve a stable umbilical for the selected lateral stability criterion (outer diameter = as specified in Umbilical data). The result is fully iterated.

### 4.2.2 Graphical Results

There are two graphs presented in the Main sheet, both are updated automatically when pushing the Calculate button, as shown in Figure 4-13, Figure 4-14 and Figure 4-15.

Figure 4-13 is found to the left of the numerical results. This graph is very illustrative, showing which wave and current direction that gives the largest weight requirements.

Figure 4-14 is found to the right of the numerical results for Pipeline mode. The concrete density interval is a set range between 1950 and 3450 kg/m$^3$ taken from experience in several on-bottom stability projects to cover all possible densities.

For umbilical mode required weight is plotted as a function of outer diameter, where the outer diameter range is set to +/- 5% relative to the outer diameter specified.

These graphs are helpful for understanding how sensitive the required submerged weight is with respect to concrete coating density (Pipeline) and outer diameter (Umbilical), when deciding on which coating or diameter to use.

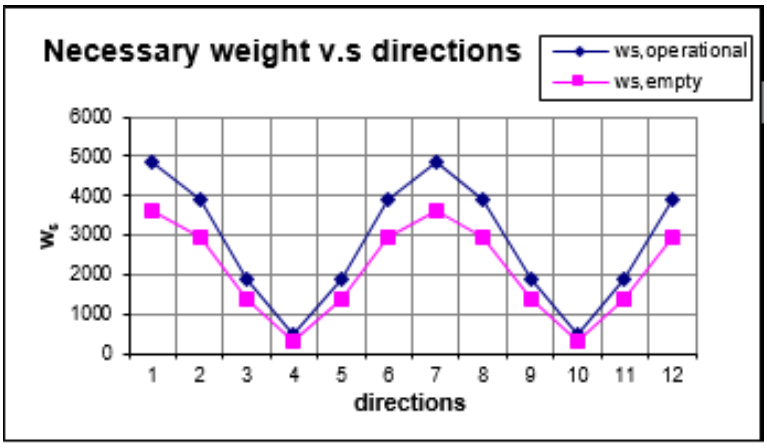



**Figure 4-13 Required weight versus wave and current direction.**

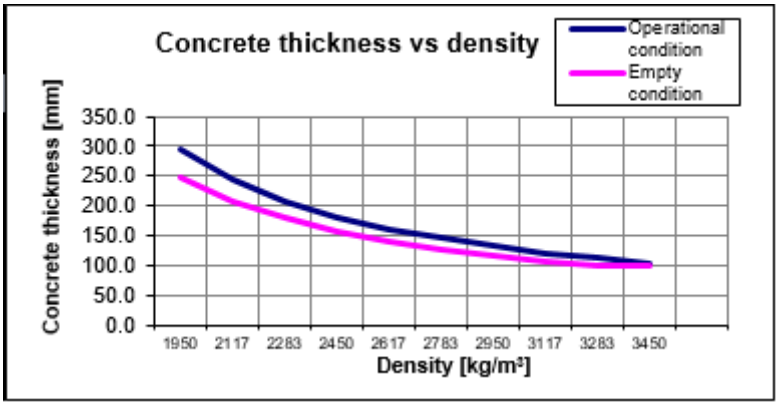



**Figure 4-14 Necessary concrete thickness for varying density, pipeline mode.**

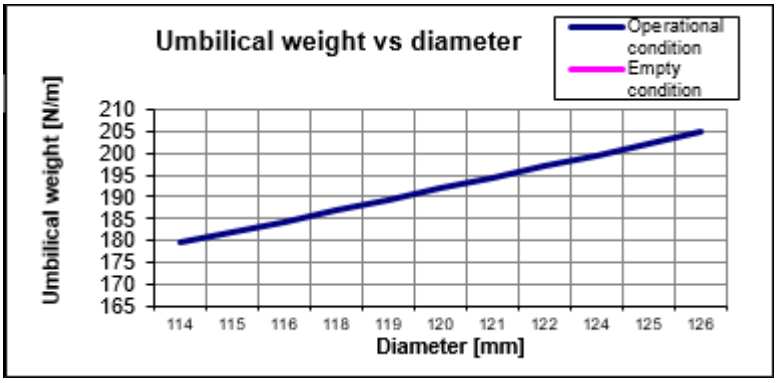



**Figure 4-15 Required weight for varying outer diameters, umbilical mode.**

### 4.2.3 Report Sheet

The Report Sheet list final results as well as relevant intermediate results and input parameters. The Report sheet is formatted so that it may be copied directly into a report's appendix or be exported as a PDF file. Example sheets for both Pipeline and Umbilical modes are found in Appendix A.

### 4.2.4 Plots Sheet

This worksheet includes additional helpful results and plots.

As described in Section 4.2.1, StableLines reports results from the worst RPV combination. Results for both RPV combinations are reported in the Plots sheet, as shown in Figure 4-16.

| Results - All RPV Combinations | | | | |
|---|---|---|---|---|
| RPV combination | 1-year Current and 10-year Wave | 10-year Current and 1-year Wave | 10-year Current and 100-year Wave | 100-year Current and 10-year Wave |
| $w_s$ [N/m] | 2329 | 2329 | 3173 | 3173 |
| $w_{necessary}$ [N/m] | 3602 | 2922 | 4867 | 3872 |

**Figure 4-16 Results for all four RPV combinations when regarding pipeline (umbilical has only operation).**

When performing calculations with StableLines, the current velocity is varied between zero and maximum design value, *V*. This is done for all given direction combinations of current and wave introduced. Figure 4-17 shows an excerpt from the Plot sheet where the necessary weight is presented as a function of current velocity.

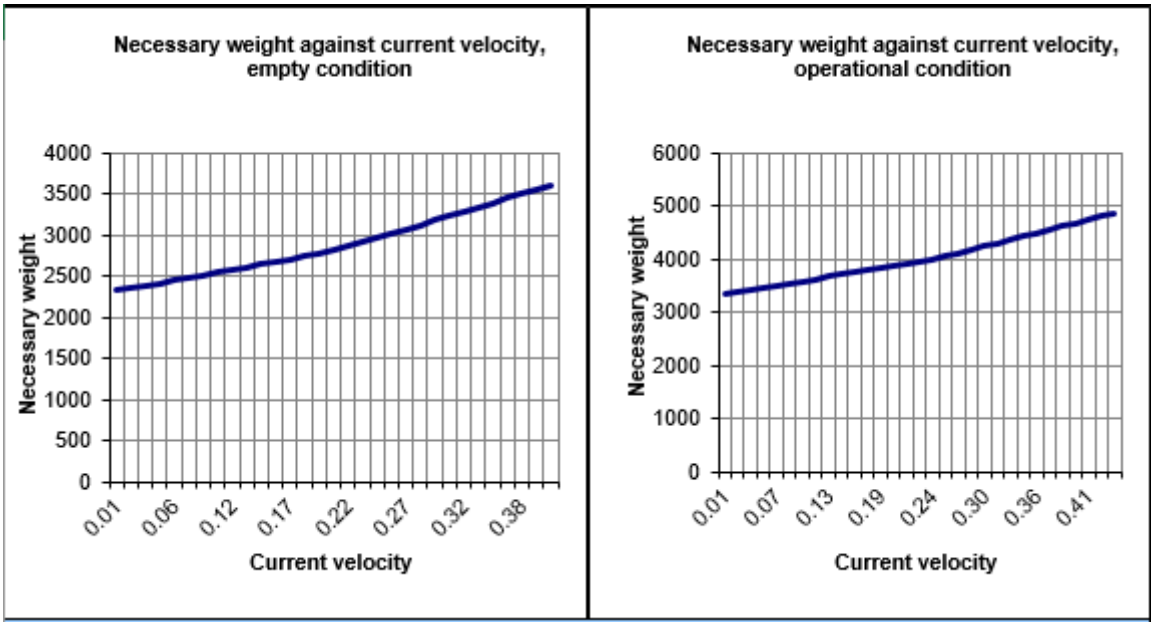



**Figure 4-17 Necessary weight as a function of current velocity for empty and operational condition.**

There are also two graphs presenting the sensitivity of the stability parameter *L* versus the steady to oscillatory velocity ratio *M* and the Keulegan-Carpenter number *K*. These two graphs are presented in Figure 4-18.

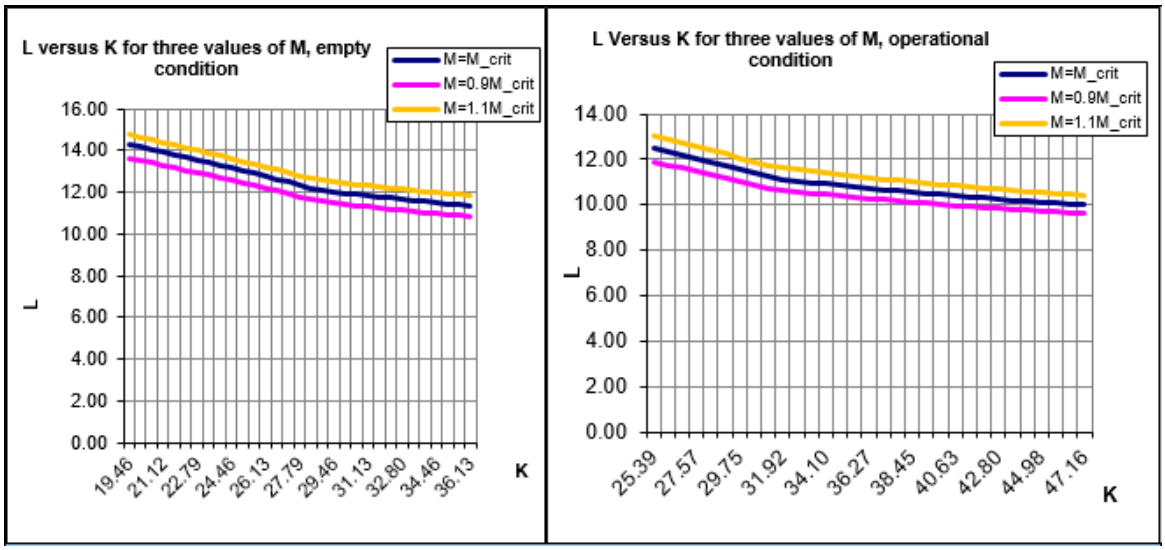



**Figure 4-18 Sensitivity of L when varying K and M for empty and operational conditions**

The critical value of *K* and *M* are retrieved from the case presented in the main sheet, i.e. the value in the middle of the *K*-interval on the blue line is the same case as presented in the main sheet and the report sheet.

The data that forms the basis for two graphs shown on the Main sheet is presented in Figure 4-19.

| Necessary weight for all directions | | | | | Concrete thickness for various concrete densities | | |
|---|---|---|---|---|---|---|---|
| Dir. combination | Wave dir. | Current dir. | Empty | Operational | Density | Empty condition | Operational condition |
| 1 | 0 | 0 | 3602 | 4867 | 1950 | 245.745 | 293.034 |
| 2 | 30 | 30 | 2964 | 3891 | 2117 | 206.326 | 242.397 |
| 3 | 60 | 60 | 1364 | 1886 | 2283 | 178.370 | 207.167 |
| 4 | 90 | 90 | 298 | 463 | 2450 | 157.204 | 180.969 |
| 5 | 120 | 120 | 1364 | 1886 | 2617 | 140.782 | 160.915 |
| 6 | 150 | 150 | 2964 | 3891 | 2783 | 127.544 | 144.838 |
| 7 | 180 | 180 | 3602 | 4867 | 2950 | 116.562 | 131.877 |
| 8 | 210 | 210 | 2964 | 3891 | 3117 | 107.450 | 121.087 |
| 9 | 240 | 240 | 1364 | 1886 | 3283 | 100.000 | 111.887 |
| 10 | 270 | 270 | 298 | 463 | 3450 | 100.000 | 104.115 |
| 11 | 300 | 300 | 1364 | 1886 | | | |
| 12 | 330 | 330 | 2964 | 3891 | | | |

**Figure 4-19 Input data for the direction vs. $w_{s,required}$ graph and concrete density effect graph**

When the absolute stability criterion is used in pipeline mode, StableLines also presents the results of an iteration to obtain the optimal required pipeline weight and concrete thickness (this feature is not used in umbilical mode), see Figure 4-20. The required weight determined for the initial pipeline configuration is presented in the main sheet as $w_{s,required}$. However, when steel or concrete coating is added, both the weight of the pipeline and the diameter change. This affects both hydrodynamic loading and soil resistance. Thus, the new pipeline configuration will give a different required weight than the initial configuration. Iteration is therefore required until convergence between $w_s$ and $w_{s,required}$ is obtained.

The presentation of the iteration results in the Plots sheet also includes many useful intermediate results, such as load coefficients and initial penetration.

Note that a similar iteration is performed for the generalized lateral stability methods in order to account for increased diameter and weight. In the results for $t_{concrete,\ added}$ and $t_{steel,\ added}$ presented in the Main sheet the effects of increased diameter and weight are therefore taken into account.

| Iteration to obtain optimal necessary pipe weight and concrete thickness for the absolute stability criterion | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Iteration no | $w_s$ [N/m] | $w_{necessary}$ [N/m] | $t_{conc}$ [m] | $C^*_Y$ | $C^*_Z$ | $F_R$ [N/m] | $z_{pi}$ [m] | $r_{tot,y}$ |
| | | | | | | | | |
| | Empty Condition: | | | | | | | |
| | | | | | | | | |
| 1 | 2329.4 | 10888.9 | 0.0252 | 2.093 | 1.953 | 276.5 | 0.019 | 0.956 |
| 2 | 604.6 | 8222.8 | 0.1208 | 1.260 | 1.075 | 0.0 | 0.007 | 0.978 |
| 3 | 2895.6 | 11801.0 | 0.2806 | 2.164 | 2.024 | 343.8 | 0.022 | 0.951 |
| 4 | 8519.9 | 18741.4 | 0.4503 | 2.548 | 2.400 | 1056.9 | 0.052 | 0.925 |
| 5 | 16947.7 | 26713.3 | 0.5916 | 2.009 | 2.129 | 2269.5 | 0.091 | 0.902 |
| 6 | 25898.7 | 31585.0 | 0.6665 | 2.183 | 2.414 | 3733.7 | 0.131 | 0.884 |
| 7 | 31347.4 | 34761.9 | 0.7091 | 2.318 | 2.619 | 4710.7 | 0.156 | 0.874 |
| 8 | 34673.3 | 36175.4 | 0.7461 | 2.385 | 2.722 | 5338.4 | 0.171 | 0.869 |
| 9 | 37689.7 | 37138.7 | 0.7331 | 2.439 | 2.804 | 5928.1 | 0.184 | 0.864 |
| 10 | 36611.0 | 36827.4 | 0.7396 | 2.420 | 2.775 | 5715.0 | 0.180 | 0.865 |
| 11 | 37148.5 | 36987.0 | 0.7363 | 2.430 | 2.790 | 5820.8 | 0.182 | 0.864 |
| 12 | 36873.6 | 36906.6 | 0.7379 | 2.425 | 2.782 | 5766.6 | 0.181 | 0.865 |
| 13 | 37010.9 | 36947.1 | 0.7375 | 2.427 | 2.786 | 5793.7 | 0.181 | 0.865 |
| 14 | 36978.3 | 36937.5 | 0.7371 | 2.427 | 2.785 | 5787.3 | 0.181 | 0.865 |
| 15 | 36940.6 | 36926.4 | 0.7365 | 2.426 | 2.784 | 5779.8 | 0.181 | 0.865 |
| 16 | 36896.9 | 36913.5 | 0.7368 | 2.425 | 2.783 | 5771.2 | 0.181 | 0.865 |
| 17 | 36918.7 | 36919.9 | 0.7370 | 2.426 | 2.784 | 5775.5 | 0.181 | 0.865 |
| | | | | | | | | |
| | Operational Condition: | | | | | | | |
| | | | | | | | | |
| 1 | 3173.2 | 20897.0 | 0.0310 | 1.327 | 1.148 | 372.8 | 0.024 | 0.944 |
| 2 | 1563.9 | 16747.9 | 0.1285 | 1.227 | 1.044 | 0.0 | 0.014 | 0.957 |
| 3 | 3960.8 | 23638.7 | 0.3754 | 1.833 | 1.679 | 469.0 | 0.028 | 0.940 |
| 4 | 13761.8 | 49806.5 | 0.8122 | 2.523 | 2.371 | 1801.1 | 0.077 | 0.907 |
| 5 | 44218.6 | 95743.1 | 1.2912 | 2.227 | 2.482 | 7323.7 | 0.216 | 0.851 |
| 6 | 96873.1 | 116722.8 | 1.4439 | 2.734 | 3.215 | 21477.7 | 0.479 | 0.775 |
| 7 | 117909.8 | 104843.3 | 1.3457 | 2.836 | 3.366 | 28744.0 | 0.595 | 0.747 |
| 8 | 104146.2 | 113743.6 | 1.3948 | 2.773 | 3.272 | 23885.7 | 0.519 | 0.765 |
| 9 | 110922.0 | 109881.2 | 1.3584 | 2.806 | 3.320 | 26228.1 | 0.556 | 0.756 |
| 10 | 105883.9 | 112851.3 | 1.3766 | 2.782 | 3.285 | 24477.3 | 0.528 | 0.763 |
| 11 | 108388.4 | 111445.4 | 1.3857 | 2.794 | 3.303 | 25341.0 | 0.542 | 0.759 |
| 12 | 109651.6 | 110683.3 | 1.3903 | 2.800 | 3.312 | 25781.6 | 0.549 | 0.758 |
| 13 | 110285.9 | 110287.2 | 1.3904 | 2.803 | 3.316 | 26004.1 | 0.553 | 0.757 |

**Figure 4-20 Iteration to obtain optimal required pipeline weight and concrete thickness for the absolute stability criterion**

## 4.3 Parametric runs

### 4.3.1 General

This option is used to calculate several on-bottom stability cases in one run. Thus, it is a very powerful tool that can be used for screening purposes, to perform sensitivity studies or just to analyse many separate cases in one run and keep the input and results together in one data sheet. For example, different KP's along a pipeline with varying input such as depth, soil and environment.

### 4.3.2 Functionality

A new Parametric runs sheet can be generated by clicking the *Create new Parametric runs* button on the Main sheet, ref. Section 4.1.2. The new sheet will be generated, only containing headings and buttons. Input values can be filled into the empty cells on the first empty row with the *Input from Main* button. If a Parametric runs sheet already exists, it will be overwritten by the newly generated one. To avoid losing results, the old Parametric runs worksheets should be renamed before generating a new sheet.

Results are presented to the left on the row. All input parameters then continue the row to the right starting with pipeline data and ending with wave and current data for all directions. One row in the Parametric run sheet corresponds to one calculation in the Main sheet. Use the *Generate runs* button, or simply copy rows to add calculation cases.

Kilometre point, KP, is located in the first column, before the results columns and Errors & Warnings column. The KP listing variable introduces the opportunity to retrieve an overview for a whole pipeline by

implementing all important KP's and its relevant input. Errors and Warnings introduce the possibility to see which cases that did not go through, or is outside the validity of the variables used.

All input parameters with white background is affecting the case studied and can be changed. All variables with light yellow background can still be changed, but will not affect the calculations.

A calculation example with varying water depth has been introduced to illustrate typical Parametric runs sheet usage. In this example, also figures presenting results and input parameters are implemented for illustration. To view the Parametric runs calculation example, see Appendix B.

Both "Calculate γ with RP-F109" and "Optimize" options are chosen by ticking them off in the main sheet. The options are chosen for all analyses, i.e. either active or inactive for all analysis in parametric runs sheet.

All Excel built-in tools, such as Charts and Cell Formatting, can be used in the Parametric runs sheet.

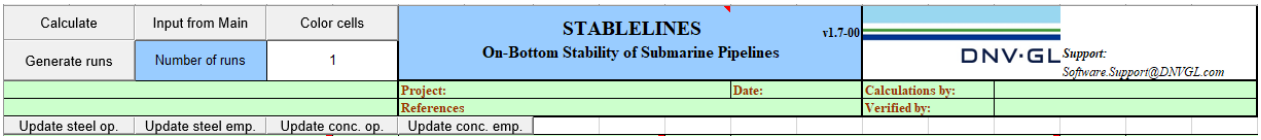



**Figure 4-21 Parametric runs heading area**

## 4.3.3 Buttons

*Calculate* The calculation routines in the parametric runs sheet are initiated using this Calculate button. Results for all sensitivity cases (rows) are calculated and updated when the Calculate button is clicked.

*Generate Runs* This button generates a number of rows/cases specified by the user (in the "number of runs" cell) by copying the input on the first row and adding the same data on consecutive rows in the Parametric runs sheet.

To generate more cases, it is also possible for the user to copy the first row of data as many times as required. The data can then be edited accordingly. Clicking the Calculate button will run the calculation for all the on-bottom stability cases, and the results are given in the sheet.

Note: If cases/rows exists from before, these will be overwritten with data copied from the first row.

*Input from Main* This button transfers all input data from Main sheet into the first input row in the Parametric runs sheet. If the first input row contains data, this data will be overwritten.

*Color cells* This button colours the background of the input parameters in accordance with the main sheet i.e. either light yellow or white. White cells are input that are included in the calculations for the respective case while light yellow background is for the input that does not affect the respective case. It also introduces a red colored background, if an input is not allowed.

*Update steel op* This button updates the steel thickness and outside diameter with the steel thickness necessary for achieving a stable pipeline during operation, ref. Section 4.2.1. This requires that a calculation has been performed to have a result to use for update. Similar buttons also exists for empty condition for steel and both operation and empty condition for concrete thickness. These buttons do not have any function in umbilical mode.

## 4.4 Notes to specific gravity

### 4.4.1 Warning messages for specific gravity

For design cases where the pipeline's/umbilical's specific gravity violates the limits stated in DNVGL-RP-F109, StablesLines will warn the user through a warning message, ref. Section 3.4. This particular warning message includes the value of $S_g$ that triggered the warning message. Often this value differs from what is reported on the Main sheet. The reason for this is that the value reported on the Main sheet is calculated based on the final report submerged weight (normally interpolated either on number of waves, duration of design storm or required displacement), while the value of $S_g$ in the warning message is based on intermediate results taken directly from the DNVGL-RP-F109 design tables.

It is the design tables in DNVGL-RP-F109 that governs the validity range of $S_g$.

### 4.4.2 Umbilicals and specific gravity

Specific gravity is defined in DNVGL-RP-F109 as $S_g = \frac{w_s+b}{b}$, where $w_s$ is submerged weight and $b$ is buoyancy.

Determining $S_g$ for umbilicals and cables is not always straight forward, as these are often permeable to seawater, which can make it tricky to determine submerged weight (from dry weight) and buoyancy.

In order to establish the applicability of the Generalized Stability criteria, ref. Section 4.4.1, StableLines currently assumes that the Umbilical is not permeable and calculates $S_g$ according to equation (3.1) in DNVGL-RP-F109, as for pipelines. The applicability of the Generalized Method is limited to $1.1 \leq S_g \leq 3.0$.

# 5 REFERENCES


/1/ DNVGL-RP-F109, *On-Bottom Stability Design of Submarine Pipelines*, May 2017

/2/ DNV-RP-F109, *On-Bottom Stability Design of Submarine Pipelines*, October 2010

/3/ DNVGL-ST-F101, "Submarine pipeline systems", October 2017

/4/ DNV-RP-C205, *Environmental Conditions and Environmental Loads*, April 2014

# APPENDIX A
# Report Sheet

A typical report sheet for Pipeline mode is shown in Figure A - 1. The most relevant input and final results are presented. In addition, intermediate results are shown for the non-dimensional on-bottom stability parameters $K$, $M$, $N$, $L$, $F_R$, $C_Y^*$, $C_Z^*$, $U_s$, $H_s$, $T_p$,$T_u$, critical RPV combo and direction as well as critical and maximum current velocity. Note that the non-dimensional parameters are reported based on $w_{s,required}$, not $w_s$.

DNV·GL

header 1
header 2
header 3
header 4

| Prepared: | Date: | Sign: | Verified: | Date: | Sign: |
|---|---|---|---|---|---|

**RELEVANT INPUT PARAMETERS:**

| **Pipeline data** | | | | | |
|---|---|---|---|---|---|
| Nominal outer steel diameter: | $D_s$ | 0.4 m | Density content: | $\rho_{cont}$ | 800 kg/m$^3$ |
| Direction of pipeline: | $\theta_{pipe}$ | 90 deg. | Cut back length FJC: | $l_{FJC}$ | 0 m |
| Density water: | $\rho_{water}$ | 1025 kg/m$^3$ | Density FJC material: | $\rho_{FJC}$ | 0 kg/m$^3$ |
| Steel thickness: | $t_{steel}$ | 0.0150 m | Steel density: | $\rho_{steel}$ | 7850 kg/m$^3$ |
| Concrete thickness: | $t_{conc}$ | 0.1000 m | Concrete density: | $\rho_{conc}$ | 2450 kg/m$^3$ |
| Marine growth thickness: | $t_{mgrt}$ | 0.0000 m | Marine growth density: | $\rho_{mgrt}$ | 1350 kg/m$^3$ |
| **Auxilliary coating layer thicknesses** | | | **Coating layer densities** | | |
| | $t_{coat,1}$ | 0.0000 m | | $\rho_{coat,1}$ | 0 kg/m$^3$ |
| | $t_{coat,2}$ | 0.0000 m | | $\rho_{coat,2}$ | 0 kg/m$^3$ |
| | $t_{coat,3}$ | 0.0000 m | | $\rho_{coat,3}$ | 0 kg/m$^3$ |
| | $t_{coat,4}$ | 0.0000 m | | $\rho_{coat,4}$ | 0 kg/m$^3$ |
| | $t_{coat,5}$ | 0.0000 m | | $\rho_{coat,5}$ | 0 kg/m$^3$ |
| **Soil data** | | | **Environmental data** | | |
| Bottom roughness param. : | $z_0$ | 5.00E-06 m | Spectral spreading exp.: | s | 8 |
| Trench depth: | $z_t$ | - m | Ref. height for current: | $z_r$ | 3 m |
| Additional penetration | $z_p$ | - m | Water depth: | d | 100 m |
| Total load reduction, horizontal: | $r_{tot,y}$ | - | Trench angle: | $\theta$ | - deg. |
| Total load reduction, vertical: | $r_{tot,z}$ | - | Peak enhancement factor: | $\gamma$ | 3.30 |
| Lift reduction factor: | $r_{perm,z}$ | - | Design storm duration: | $T_{storm}$ | 3 hrs |
| Friction coefficent: | $\mu$ | - | Abs. Stab. safety fac. | $\gamma_{SC,empty}$ | - |
| Submerged unit soil weight(Sand): | $\gamma_s'$ | - N/m$^3$ | Abs. Stab. safety fac: | $\gamma_{SC,operation}$ | - |
| Dry unit soil weight(clay): | $\gamma_s$ | 18000 N/m$^3$ | Soil type: | | Clay |
| Undrained shear strength (clay): | $s_u$ | 10000 N/m$^2$ | | | |
| Soil strength parameter (clay): | $G_c$ | 0.93 | | | |
| **Stability criterion:** | | **10D Displacement** | | | |

| **INTERMEDIATE RESULTS:** | **Empty condition** | **Operational condition** |
|---|---|---|
| Critical RPV combination | 1-year Current<br>10-year Wave | 10-year Current<br>100-year Wave |
| Critical current velocity - V/V$^*$ | 0.402 | 0.442 |
| Maximum current velocity | 0.402 | 0.442 |
| Critical direction [deg] | 0 0 | 0 0 |
| Significant wave height - $H_s$ | 14.0 | 16.0 |
| Peak period - $T_p$ | 17.0 | 18.0 |
| Wave induced velocity - $U_s/U_s^*$ | 0.974 | 1.215 |
| K/K$^*$ | 27.8 | 36.3 |
| M (Critical curr. velocity)/M$^*$ | 0.413 | 0.364 |
| M (Maximum curr. velocity)/M$^*$ | 0.413 | 0.364 |
| N/N$^*$ | 0.006 | 0.007 |
| L/L$^*$ | 12.342 | 10.726 |
| $F_R$ | - | - |
| $C_Y^*$ | - | - |
| $C_Z^*$ | - | - |
| $T_s/T^*$ | 17.12 | 17.92 |
| **FINAL RESULTS:** | | |
| $w_s$ [N/m] | 2329 | 3173 |
| $w_{s,required}$ [N/m] | 3602 | 4867 |
| $t_{conc,added}$ [m] | 0.038 | 0.063 |
| $t_{steel,added}$ [m] | 0.014 | 0 |
| $w_{s,\ vertical}$ [N/m] | 284 | 284 |

**Figure A - 1 Report sheet example case, Pipeline mode**

A typical report sheet for Umbilical mode is shown in Figure A - 2.

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header 1
header 2
header 3
header 4

| Prepared: | Date: | Sign: | Verified: | Date: | Sign: |
|---|---|---|---|---|---|

**RELEVANT INPUT PARAMETERS:**

**Umbilical data**

| | | | | | |
|---|---|---|---|---|---|
| Outer diameter of umbilical: | $D_o$ | 0.12 m | Density content: | $\rho_{cont}$ | - kg/m$^3$ |
| Direction of umbilical: | $\theta_{pipe}$ | 90 deg. | Cut back length FJC: | $l_{FJC}$ | - m |
| Density water: | $\rho_{water}$ | 1025 kg/m$^3$ | Density FJC material: | $\rho_{FJC}$ | - kg/m$^3$ |
| Marine growth thickness: | $t_{mgrt}$ | 0.0000 m | Marine growth density: | $\rho_{mgrt}$ | 1350 kg/m$^3$ |

| **Auxilliary coating layer thicknesses** | | | **Coating layer densities** | | |
|---|---|---|---|---|---|
| | $t_{coat,1}$ | - m | | $\rho_{coat,1}$ | - kg/m$^3$ |
| | $t_{coat,2}$ | - m | | $\rho_{coat,2}$ | - kg/m$^3$ |
| | $t_{coat,3}$ | - m | | $\rho_{coat,3}$ | - kg/m$^3$ |
| | $t_{coat,4}$ | - m | | $\rho_{coat,4}$ | - kg/m$^3$ |
| | $t_{coat,5}$ | - m | | $\rho_{coat,5}$ | - kg/m$^3$ |

| **Soil data** | | | **Environmental data** | | |
|---|---|---|---|---|---|
| Bottom roughness param. : | $z_0$ | 5.00E-06 m | Spectral spreading exp.: | s | 8 |
| Trench depth: | $z_t$ | - m | Ref. height for current: | $z_r$ | 3 m |
| Additional penetration | $z_p$ | - m | Water depth: | d | 250 m |
| Total load reduction, horizontal: | $r_{tot,y}$ | - | Trench angle: | $\theta$ | - deg. |
| Total load reduction, vertical: | $r_{tot,z}$ | - | Peak enhancement factor: | $\gamma$ | 3.30 |
| Lift reduction factor: | $r_{perm,z}$ | - | Design storm duration: | $T_{storm}$ | 3 hrs |
| Friction coefficent: | $\mu$ | - | Abs. Stab. safety fac. | $\gamma_{SC,empty}$ | - |
| Submerged unit soil weight(Sand): | $\gamma_s'$ | - N/m$^3$ | Abs. Stab. safety fac: | $\gamma_{SC,operation}$ | - |
| Dry unit soil weight(clay): | $\gamma_s$ | 18000 N/m$^3$ | Soil type: | | Clay |
| Undrained shear strength (clay): | $s_u$ | 10000 N/m$^2$ | | | |
| Soil strength parameter (clay): | $G_c$ | 2.78 | | | |
| **Stability criterion:** | | **10D Displacement** | | | |

| **INTERMEDIATE RESULTS:** | **Empty condition** | **Operational condition** |
|---|---|---|
| Critical RPV combination | - | 10-year Current |
| | - | 100-year Wave |
| Critical current velocity - V/V* | - | 0.376 |
| Maximum current velocity | - | 0.376 |
| Critical direction [deg] | - - | 0 0 |
| Significant wave height - $H_s$ | - | 16.0 |
| Peak period - $T_p$ | - | 18.0 |
| Wave induced velocity - $U_s/U_s^*$ | - | 0.244 |
| K/K* | - | 40.6 |
| M (Critical curr. velocity)/M* | - | 1.539 |
| M (Maximum curr. velocity)/M* | - | 1.539 |
| N/N* | - | 0.001 |
| L/L* | - | 52.392 |
| $F_R$ | - | - |
| $C_Y^*$ | - | - |
| $C_Z^*$ | - | - |
| $T_y/T^*$ | - | 19.95 |
| **FINAL RESULTS:** | | |
| $w_{s,required}$ [N/m] | - | 192 |

**Figure A - 2 Report sheet example case, Umbilical mode**

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# APPENDIX B
# Parametric Runs Example Case

This Parametric runs calculations example is set up to illustrate a typical Parametric runs sheet usage. A whole pipeline section is to be investigated for the 0.5D Criterion. Therefore, critical KP's are defined by the user directly in the sheet to retrieve full overview. As can be seen in Figure B - 1, KP values are introduced to the very left of the sheet.

| Initial penetration for water-filled pipe | | | | | FALSE | Initial penetration for water-filled pipe | | | | FALSE | Boundary layer correction for current | | | | TRUE | Calculate necessary thickness | | | | | | TRUE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Design condition empty condition<br>1-year and 10-year RPV Combination | | | | | | Design condition operational condition<br>10-year and 100-year RPV Combination | | | | | | Pipeline data | | | | | | | | | | |
| KP | $w_s$ [N/m] | $w_{s,required}$ [N/m] | $t_{conc,added}$ [m] | $t_{steel,added}$ [m] | $w_{s,vertical}$ [N/m] | $w_s$ [N/m] | $w_{s,required}$ [N/m] | $t_{conc,added}$ [m] | $t_{steel,added}$ [m] | $w_{s,vertical}$ [N/m] | Errors & Warnings | Mode | $\theta_{pipe}$ [deg] | $D_s$ [m] | $t_{steel}$ [m] | $t_{conc}$ [m] | $t_{mgrt}$ [m] | $t_{FJC}$ [m] | $\rho_{water}$ [kg/m$^3$] | $\rho_{steel}$ [kg/m$^3$] | $\rho_{conc}$ [kg/m$^3$] | $\rho_{mgrt}$ [kg/m$^3$] | $\rho_{cont}$ [kg/m$^3$] |
| 0 | 2329 | 3748 | 0.0883 | 0.0172 | 284 | 3173 | 5229 | 0.1548 | 0.0257 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 23 | 2329 | 2666 | 0.0234 | 0.0041 | 284 | 3173 | 3704 | 0.0513 | 0.0066 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 46 | 2329 | 2149 | -0.0109 | -0.015 | 284 | 3173 | 2807 | -0.0264 | -0.0046 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 69 | 2329 | 1797 | -0.0316 | -0.0064 | 284 | 3173 | 2438 | -0.0556 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 92 | 2329 | 1446 | -0.0506 | -0.0106 | 284 | 3173 | 1973 | -0.0821 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 115 | 2329 | 1140 | -0.0653 | -0.015 | 284 | 3173 | 1598 | -0.1 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 138 | 2329 | 862 | -0.076 | -0.015 | 284 | 3173 | 1300 | -0.1 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 161 | 2329 | 657 | -0.0842 | -0.015 | 284 | 3173 | 1017 | -0.1 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 184 | 2329 | 502 | -0.0894 | -0.015 | 284 | 3173 | 804 | -0.1 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |
| 207 | 2329 | 415 | -0.0925 | -0.015 | 284 | 3173 | 634 | -0.1 | -0.015 | 284 | | Pipe | 90 | 0.4 | 0.0150 | 0.1000 | 0.0000 | 0.0000 | 1025 | 7850 | 2450 | 1350 | 800 |

**Figure B - 1 KP values varies between 2 and 141 for this particular case**

The only parameter changed in this parametric study is the water depth as shown in Figure B - 2. This is one of the environmental parameters, thus this whole section is shown. Note that the water depth increases with increasing KP values.

| Environmental Parameters | | | | | |
|---|---|---|---|---|---|
| s | $z_r$ [m] | d [m] | $\theta_t$ [deg] | $\gamma$ | $T_{storm}$ [hrs] |
| 8 | 3 | 80 | - | 3.3 | 3 |
| 8 | 3 | 100 | - | 3.3 | 3 |
| 8 | 3 | 120 | - | 3.3 | 3 |
| 8 | 3 | 140 | - | 3.3 | 3 |
| 8 | 3 | 160 | - | 3.3 | 3 |
| 8 | 3 | 180 | - | 3.3 | 3 |
| 8 | 3 | 200 | - | 3.3 | 3 |
| 8 | 3 | 220 | - | 3.3 | 3 |
| 8 | 3 | 240 | - | 3.3 | 3 |
| 8 | 3 | 260 | - | 3.3 | 3 |

**Figure B - 2 The environmental parameter d, water depth, is varied along the pipeline length**

The results for the varying water depths are shown in the Figure B - 3.

| Initial penetration for water-filled pipe | | | | | FALSE | Initial penetration for water-filled pipe | | | | FALSE |
|---|---|---|---|---|---|---|---|---|---|---|
| Design condition empty condition | | | | | | Design condition operational condition | | | | |
| 1-year and 10-year RPV Combination | | | | | | 10-year and 100-year RPV Combination | | | | |
| KP | $w_s$ [N/m] | $w_{s,\ required}$ [N/m] | $t_{conc,added}$ [m] | $t_{steel,added}$ [m] | $w_{s,\ vertical}$ [N/m] | $w_s$ [N/m] | $w_{s,\ required}$ [N/m] | $t_{conc,added}$ [m] | $t_{steel,added}$ [m] | $w_{s,\ vertical}$ [N/m] |
| 0 | 2329 | 3748 | 0.0883 | 0.0172 | 284 | 3173 | 5229 | 0.1548 | 0.0257 | 284 |
| 23 | 2329 | 2666 | 0.0234 | 0.0041 | 284 | 3173 | 3704 | 0.0513 | 0.0066 | 284 |
| 46 | 2329 | 2149 | -0.0109 | -0.015 | 284 | 3173 | 2807 | -0.0264 | -0.0046 | 284 |
| 69 | 2329 | 1797 | -0.0316 | -0.0064 | 284 | 3173 | 2438 | -0.0556 | -0.015 | 284 |
| 92 | 2329 | 1446 | -0.0506 | -0.0106 | 284 | 3173 | 1973 | -0.0821 | -0.015 | 284 |
| 115 | 2329 | 1140 | -0.0653 | -0.015 | 284 | 3173 | 1598 | -0.1 | -0.015 | 284 |
| 138 | 2329 | 862 | -0.076 | -0.015 | 284 | 3173 | 1300 | -0.1 | -0.015 | 284 |
| 161 | 2329 | 657 | -0.0842 | -0.015 | 284 | 3173 | 1017 | -0.1 | -0.015 | 284 |
| 184 | 2329 | 502 | -0.0894 | -0.015 | 284 | 3173 | 804 | -0.1 | -0.015 | 284 |
| 207 | 2329 | 415 | -0.0925 | -0.015 | 284 | 3173 | 634 | -0.1 | -0.015 | 284 |

**Figure B - 3 Parametric runs results for all KP values**

The necessary concrete thickness versus water depth for empty and operational condition is shown in Figure B - 4. The results are found to be as expected. As mentioned above, water depth is increased downwards. The needed concrete thickness and added steel thickness decrease and is zero for larger water depths. Required submerged weight also decreases for increasing water depth. This provides the user the possibility to suggest different measures to obtain stability for the different sections of the pipeline. For instance, it is noticed that the pipeline in operational condition is stable without adding steel or concrete for water depths over 180m. Note that for this particular case, the Optimize check box is hooked off, ref. Section 4.2.1.

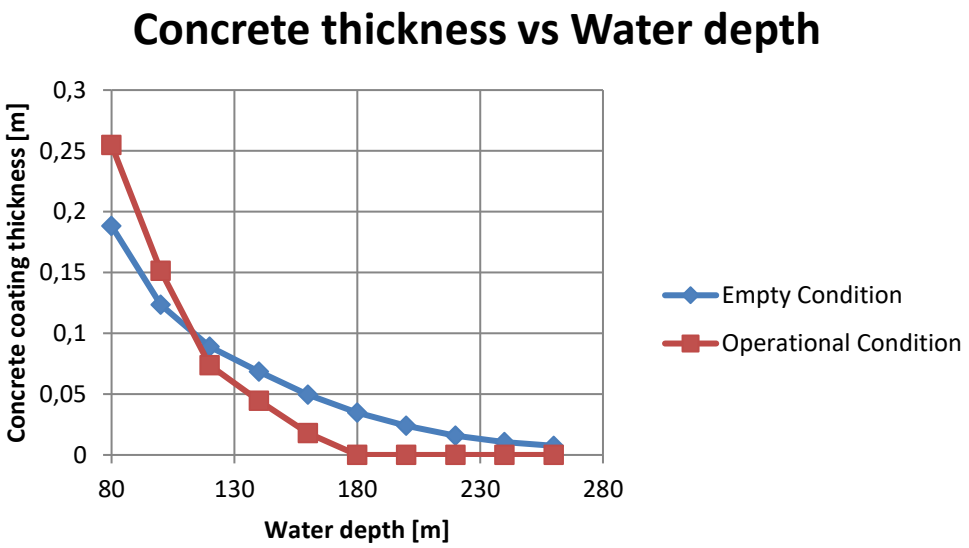



**Figure B - 4 Necessary concrete thickness vs. water depth for empty and operational condition.**

As illustrated above, the parametric runs sheet is an illustrative and useful tool. StableLines provides the possibility of performing several on-bottom stability calculations at the same time, reducing the time for analyses by a significant factor.

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